Subject Code : 23A01703a

GROUND IMPROVEMENT TECHNIQUES

UNIT - I

Expansive Soils: Problems, Identification, Swell Pressure, Improvement and Under-reamed Piles

Syllabus topics covered in this unit

- Problems of Expansive Soils
- Tests for Identification of Expansive Soils
- Methods of Determination of Swell Pressure
- Improvement of Expansive Soils
- Foundation Techniques in Expansive Soils
- Under-reamed Piles

How to use these notes

- Red words = keywords to remember for the exam.
- Green words = examples, units and values.
- Yellow boxes = quick points; pink boxes = important questions / formulae.
- Learn the sketches - they fetch marks in the answer paper.

1. Expansive Soils - Introduction

Expansive soils are fine grained clayey soils that show large volume change - they swell on absorbing water and shrink on losing water. The movement is not caused by load but by change in moisture content. They are found in arid and semi-arid regions where evaporation is greater than rainfall.
In India these are the Black Cotton (BC) soils (regur), covering about 20 percent of the land - Deccan plateau, Maharashtra, Madhya Pradesh, Gujarat, Karnataka and Andhra Pradesh (Rayalaseema, Kurnool, Anantapur, Guntur, Krishna). They are rich in lime, iron and magnesia, and dark in colour because of titaniferous magnetite and humus.

Why do they swell? - clay mineralogy

Clay mineral	Structure	Swelling	Specific surface (m^2/g)
Montmorillonite (smectite)	2:1 layer (silica-alumina-silica), weak oxygen-oxygen bond	Very high; water enters between the layers	800
Illite	2:1 layer with potassium bond	Medium	80
Kaolinite	1:1 layer with hydrogen bond	Low	15

- Montmorillonite is the main cause: its sheets are bonded weakly so water and exchangeable cations enter between them and push the sheets apart (interlayer swelling).
- Diffuse double layer and osmotic repulsion adsorb more water when the electrolyte concentration is low.
- Soil particle is flat and negatively charged; it attracts cations (Na^+, Ca^{2+}) and dipolar water. Sodium montmorillonite swells more than calcium montmorillonite.

Factors affecting swelling

- Type and amount of clay mineral (montmorillonite %), cation exchange capacity.
- Initial water content - dry soil swells more; dry density - denser soil swells more.
- Soil structure - flocculated vs dispersed; compaction method (static / dynamic).
- Surcharge - higher load reduces swelling; drainage and climate - wet-dry cycle, vegetation (trees remove moisture), leaking pipes, water table fluctuation.
- Time - swelling is time dependent, mostly in the first monsoon seasons after construction.

2. Problems of Expansive Soils

The seasonal moisture change in the top active zone (depth 1.5 to 3.5 m in India, up to 5 m or more in arid areas) causes cyclic heave and shrinkage and repeated movement of foundations and structures. The depth of the active zone is the depth up to which seasonal moisture variation takes place.

Effects on structures

Structure	Typical damage
Light buildings (1 - 2 storeys), compound walls	Diagonal and vertical cracks in walls, door and window frames jammed, floors cracked and heaved, differential settlement and tilting; "doming" in the centre and edge shrinkage in dry period
Pavements, runways, canal lining	Longitudinal cracking, waves and undulations, edge failure, uplift and breaking of lining, seepage loss
Retaining walls, basements	Lateral swelling pressure larger than active earth pressure; tilting, cracking
Pipelines, culverts, service lines	Breakage, leakage at joints which then wets the soil further
Earth dams, embankments, cuttings	Slope failure (shallow slides), cracks from desiccation
Deep foundations - piles	Uplift (jacking) of the pile by adhesion in the active zone, loss of contact
Transmission towers, tanks	Tilting, uplift of footings

- Crack pattern: in the dry period cracks start at the corners and wide at the top (edge shrinkage); in the wet period the centre heaves (centre lift) - cracks wide at the bottom.
- Greatest damage to lightly loaded structures because the swelling pressure (50 - 500 kPa and more) is greater than the structural load.
- Movement is non-uniform - near the walls the soil is shielded from evaporation and stays moister than the soil in the open, so edge and centre differ.
- Vegetation (trees) near buildings dries the soil locally and causes shrinkage settlement.
- The annual loss in India due to damage of structures on expansive soil runs to thousands of crores of rupees.

Remember

- Swell = heave on wetting; shrinkage = settlement and cracks on drying; the cycle repeats every year.
- Expansive soil is not weak (high strength when dry) - the problem is volume change, not bearing capacity.

3. Tests for Identification of Expansive Soils

Expansive soils are identified by field indications, index property tests and direct swell tests. Mineralogical tests give the clay type.

Field identification

- Dark grey / black colour, deep wide cracks (polygonal pattern) in summer, sticky and plastic in the monsoon, soap-like feel.
- Local history of building damage; presence of calcareous nodules (kankar) in lower layers.

Laboratory tests

Test	Method (IS)	Remarks
Free swell index (FSI)	IS 2720 Part 40: 10 g oven dried soil passing 425 micron into two 100 ml graduated jars - one with water, one with kerosene; read volume after 24 h	$FSI = (V_d - V_k) / V_k \times 100$ %
Atterberg limits	Liquid limit (LL), plastic limit (PL), shrinkage limit (SL), plasticity index PI = LL - PL	High LL and PI -> high swell; low SL
Colloid content / clay size fraction	Hydrometer analysis, % finer than 0.002 mm	Higher clay -> more swelling
Activity (Skempton)	A = PI / (% clay fraction < 2 micron)	A > 1.25 active; montmorillonite A = 1 - 7
Linear shrinkage	Shrinkage of bar in a mould	Used with LL
Cation exchange capacity (CEC)	Ammonium acetate method	Montmorillonite: 80 - 150 meq/100 g
Mineralogical analysis	X-ray diffraction (XRD), DTA, electron microscope, SEM	Direct identification of the mineral

Free swell index (IS 1498 / IS 2720-40)

- $FSI\ (\%) = (V_d - V_k) / V_k \times 100$; V_d = volume in distilled water, V_k = volume in kerosene (non polar, no swelling).
- Example: V_d = 18 cc, V_k = 10 cc => FSI = 80 percent (high degree of expansiveness).

Classification by degree of expansion (IS 1498)

Free swell index (%)	Degree of expansion	Probable damage
< 20	Low	Negligible
20 - 35	Moderate	Some damage
35 - 50	High	Considerable
> 50	Very high	Severe

Classification by index properties (Holtz and Gibbs / Chen)

LL (%)	PI (%)	SL (%)	Colloid (%)	Probable expansion (% - dry soil at 7 kPa)	Degree
20 - 35	< 18	> 15	< 15	< 10	Low
35 - 50	15 - 28	10 - 15	13 - 23	10 - 20	Medium
50 - 70	25 - 41	7 - 12	20 - 31	20 - 30	High
70 - 90	> 35	< 11	> 28	> 30	Very high

- Chen (1988) by PI: 0 - 15 low; 10 - 35 medium; 20 - 55 high; > 35 very high potential for swell.
- Seed, Woodward and Lundgren (1962) relate swell potential S to activity and clay fraction: $S = 3.6 \times 10^{-5} A^{2.44} C^{3.44}$ (C = clay %, A = activity).
- Differential free swell (DFS) of the soil is also used by Indian engineers: DFS < 20 = low and > 50 = very high.
- Direct test: one dimensional oedometer swell test - swell % = change in height / original height x 100 under the seating load.

4. Determination of Swell Pressure

Swell pressure is the pressure that must be applied to prevent any volume increase of a soil sample when it is given free access to water. Swelling pressure is the maximum pressure developed (kPa). It tells the structural load needed or the uplift force on a footing or pile.

Methods (IS 2720 Part 41 - oedometer tests)

Method	Procedure	Remarks
1. Constant volume method	Undisturbed (or remoulded) sample in the oedometer ring. Add water; as the sample tries to swell, add load to keep the dial gauge reading constant (no change in height). The final load / area = swell pressure	Direct and quick; the sample is not allowed to swell at all; tends to give the lower value for high swelling soil
2. Swell-consolidation (loading after swell) method	Let the sample swell freely under seating load, then load it in the usual increments until it is compressed back to its original height. The pressure at the original void ratio e_0 = swell pressure	Gives higher value; the soil is disturbed by swelling; takes longer
3. Free-swell / Zero swell (Method of 3 or 4 specimens)	Several identical specimens are wetted under different surcharges (e.g., 6.25, 12.5, 25, 50 kPa); measure the % swell / heave in each and plot against log pressure; pressure for zero swell = swell pressure	Needs more samples; gives the swell-load relation; reliable
4. Double oedometer / Jennings & Knight	One sample at natural moisture and another soaked are loaded; compare the e - log p curves	Also gives collapse and heave prediction

Constant volume method - steps

Trim sample, fit ring, saturate porous stones → Seat load 5 kPa; add water → Add load to keep height constant → Max load at equilibrium = P_s

Empirical relations

* Komornik and David (1969): $\log P_s$ (kg/cm^2) = 2.132 + 0.0208 LL + 0.000665 γ_d - 0.0269 w
* Typical swell pressure: LL 50 -> 50 - 100 kPa; LL 80 -> 300 - 600 kPa; BC soil in India: 50 - 400 kPa.
* Heave of a layer: $\Delta H = (C_s H / (1 + e_0)) \log ((\sigma_0 + \sigma_s) / \sigma_0)$ (C_s = swell index)

> Other methods: swell pressure from the swell-load curve (Fredlund), Free swell + consolidation (Sridharan), Chen's correlation with PI, and in field plate load test with water.

> Use: the dead load pressure should be more than P_s to avoid heave; if not, adopt foundation techniques or improve soil. Swell pressure acts on retaining walls and basement walls as extra lateral load.

5. Improvement of Expansive Soils

The aim of improvement is to reduce the swell potential, or to control the moisture variation, or to remove the soil from the active zone. The methods are of four groups.

A. Replacement and moisture control

- Replacement - remove expansive soil up to the active zone depth and replace it with a non-expansive granular soil. Costly when the depth is large.
- Cohesive Non-Swelling (CNS) layer (IS 9451 / Katti): a layer of 0.9 to 1.2 m (min 0.6) of non-swelling cohesive soil (PI < 20, low swell) placed below foundations, canal lining and pavements. The CNS layer gives overburden and a cushion, reduces suction, and so swell is controlled to the lower depth.
- Pre-wetting (ponding) - soil is allowed to swell before construction by ponding water for weeks; heave is completed beforehand. Slow, works for large flat sites (roads, canals).
- Moisture control - keep the moisture constant: footpath / apron (min 1.5 m wide) around the building, impermeable horizontal and vertical moisture barriers (polyethylene, bitumen, concrete), good surface drainage, plumbing leaks avoided, no trees close to the building (distance > height of the tree), avoid flower beds near the walls.
- Compaction control - compact on the wet side of OMC with low dry density (e.g., 95 percent of the standard Proctor dry density at OMC + 2 percent) to reduce swell.

B. Chemical stabilisation

Stabiliser	Action	Dose
Lime	Cation exchange (Ca^{2+} replaces Na^{+}), flocculation, pozzolanic reaction forms cementing gel -> lower PI, swell and plasticity	4 - 8 percent by weight (initial consumption of lime, ICL by pH test - Eades Grim)
Cement	Hydration cements the particles; for low PI soils	5 - 10 percent
Fly ash + lime	Pozzolanic reaction; reduces cost and swell	Lime 3-5 %, fly ash 10-20 %
Calcium chloride, sodium chloride (salt)	Reduce the double layer, hygroscopic - reduce moisture loss	1 - 3 percent
Organic cationic compounds, potassium compounds, enzymes (EKM), Terrazyme	Reduce the water affinity of clay	Manufacturer dose
Bitumen emulsion	Waterproofing the soil	Special cases

- Lime piles / lime columns: boreholes (dia 0.2 - 0.4 m) filled with quicklime or lime-sand; lime diffuses into the soil. Lime slurry pressure injection (LSPI) through boreholes up to 3 - 4 m depth; the slurry penetrates the cracks of the soil.
- Lime treatment produces a quick reduction in plasticity, but also needs curing; sulphate-bearing soils may heave on lime treatment (ettringite) - check.

C. Mechanical and physical methods

- Surcharge loading (placing an overburden to counter the swelling), dynamic compaction, thermal treatment (heating to 400 C decreases the swell by changing the clay), soil reinforcement with fibres or geosynthetics, geocell mattress, remove / sand cushion under footings.
- Prewetting + lime / CNS combination is used under canals (e.g., Nagarjuna Sagar canal, Tungabhadra).

D. Structural and foundation approach

Instead of changing the soil, the structure may be isolated from the movement - see the next section on foundation techniques.

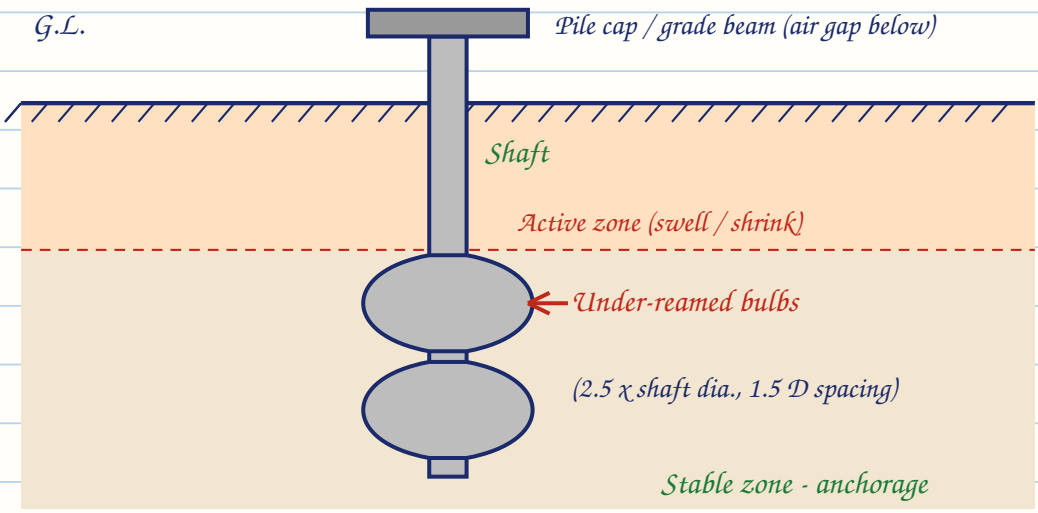
6. Foundation Techniques in Expansive Soils

The foundation is placed so that it is either below the active zone, or isolated from the soil movement, or is rigid enough to resist the movement. Common choices are the following.

Technique	Details
Deep foundation below the active zone	Drilled piers / bored piles / under-reamed piles anchored in the stable zone below the seasonal moisture change; resist uplift by their anchorage
Foundation on CNS layer / sand cushion	Strip or isolated footings on a 0.9 - 1.2 m CNS layer; shallow foundations at 1.0 - 1.5 m (but need caution); swell pressure on footing is reduced
Raft foundation / stiffened raft	Rigid RC raft or waffle raft with stiffening beams to move as one unit without bending (the structure "floats" on the moving soil); for light buildings
Isolated footing with tie beams	Footings on a sand cushion + plinth / tie beams (RC bands) at plinth, lintel and roof levels to give rigidity
Belled piers / drilled shafts	Large diameter drilled piers with bell; with void form below the grade beam
Pier and beam with void space	Grade beams spanning between piles, with a void (air gap 8 - 10 cm) below the beam so the swelling soil does not push the beam
Shallow foundation with controlled moisture	Moisture barrier around building, vertical barrier to depth of 1.5 m, apron, drain; reduce moisture changes
Structural design measures	Light weight structure, flexible joints, expansion joints and separate sections, slip joints at doors, plumbing with flexible connections, no brick work in direct contact (slip layer between plinth masonry and the beam)

- Principle: (i) Avoid the swelling zone; (ii) Increase the dead load more than the swell pressure; (iii) Make the structure rigid / flexible to match the movement; (iv) Prevent the water from reaching the soil.
- Tie beams and lintel bands in buildings. Roof level sloped for drainage, fix the plinth protection up to 1 m and no cut in the plinth by pipes.
- Retaining walls: use a granular backfill, drainage, design for swell pressure.
- Pavement and canal: CNS layer, lime treatment, prewetting, sub-surface drain, narrow shoulder with impermeable cover.

7. Under-reamed Piles



Under-reamed pile is a bored cast-in-situ concrete pile with one or more bulbs (under-reams) formed by enlarging the bore at the lower part. It was first developed in India by Central Building Research Institute (CBRI), Roorkee, for black cotton soils. IS 2911 (Part 3) covers the design and construction.

Why under-reamed piles in expansive soils?

- The pile is taken below the active zone, in the stable moisture zone, where the soil does not swell or shrink.
- The bulbs act as anchors against the uplift forces caused by the swelling soil acting on the shaft in the active zone (the soil tries to lift the shaft by skin friction).
- The bulbs also give additional bearing capacity (end bearing and annular bearing).
- Suited for light and medium loaded structures; can be built with simple hand operated equipment.

Geometry (IS 2911 Part 3)

Feature	Recommended value
Shaft diameter	20 - 50 cm (20, 25, 30, 35, 40 cm commonly)
Bulb diameter D_u	2.0 to 3.0 times the shaft diameter (usually 2.5 d for black cotton soil, 2.0 d in hard soil); 2.5 d is most used
Number of bulbs	Single or double (multi) bulb; double under-reamed piles are common
Spacing of bulbs	Not more than $1.5 \times D_u$ centre to centre; the top bulb at not less than $2\ D_u$ below the ground (usually below the active zone)
Minimum length	3.5 m below the ground level in expansive soil (not less than 3 m) and the bulbs below the active zone depth
Spacing of piles	Not less than $2 \times D_u$ (for under-reamed) and 3 d for straight shaft
Reinforcement	Minimum 0.4 % longitudinal steel (min 3 bars, 10 mm) + stirrups 6 mm at 15 cm; for uplift 0.4 to 0.8 %
Concrete	M 15 to M 20; grade beam 20 cm min

Construction steps

Bore the shaft with auger (hand / rig) → Under-ream with the cutter tool (bulbs) → Place the reinforcement cage → Pour concrete by tremie; grade beam over the pile

- Boring dry: no bentonite for BC soil; casing in soft soils; the under-reaming tool has two hinged blades that open by pushing the tool down.
- Concreting immediately after boring to prevent collapse; tremie / hopper to avoid segregation; compact by rodding.
- Grade beam (plinth beam) connects the pile heads and supports the wall; keep an air gap of 8 - 10 cm (sand / void form) between the beam bottom and the ground so that the swelling soil does not lift the beam.
- Pile top is anchored into the grade beam. Piles also under compound walls and in canal bridges.

Load carrying capacity (IS 2911 Part 3)

Ultimate load of single pile in clay

- $Q_u = A_p N_c c_p + A_u N_c c_u$ + alpha $c_s A_s$ (compression)
- A_p = base area of the bottom bulb; A_u = annular area of the other bulbs $(pi/4)(D_u^2 - d^2)$; $N_c = 9$; c_p, c_u = cohesion at bulb levels; alpha = adhesion factor (0.3 - 0.5 for BC soil, often shaft friction in the active zone is neglected); A_s = shaft surface area below the active zone.
- Safe load $Q_s = Q_u / FS$, $FS = 2.5$ (compression), and for uplift $Q_a = Q_u / 3$ in uplift.
- Under-reamed pile capacity: 25 to 80 kN (single bulb 20-30 cm shaft), 100 - 250 kN for bigger piles.

Design for uplift (swelling soil)

- Uplift force $F_u = pi\ d\ L_a\ tau_s$ where L_a = depth of the active zone and tau_s = swelling soil shear (about 0.5 c or 20 - 50 kPa).
- Check: anchorage resistance in the stable zone $Q_{anch} = c N_c A$ (bulbs) + side friction >= F_u - W (dead load) with a factor of safety about 1.0 - 1.5 on the uplift; if not, increase the length, add bulbs or increase the load by dead weight.
- Steel for the tension at the pile - usually the area $A_{st} = F_u / (0.87 f_y)$ with the minimum percentage.
- Pile length should be at least 3 m and extend above the equilibrium moisture depth; many designs take the length $L = 3.5$ to 5 m with two bulbs.

Advantages

- Simple technique, low cost, uses local labour; suitable for BC soils and also for loose granular and filled up soils; resists uplift and loads, little settlement.
- Limitations: not for soft clays with high water table (bulb collapse), special tools needed; the bulbs cannot be properly formed in loose sand below the water table.

Quick revision of Unit I

- Expansive soil: montmorillonite; BC soil; swell-shrink cycle; active zone 1.5 - 3.5 m.
- Identification: FSI, DFS, LL, PI, SL, activity; IS classification of FSI (< 20 to > 50).
- Swell pressure: constant volume, swell-consolidation, zero swell methods. Improvement: CNS layer, lime, prewetting, moisture barrier, replacement.
- Foundations: under-reamed piles (bulb = 2.5 d, spacing 1.5 D_u, min length 3.5 m, below active zone), raft, grade beams with air gap.

Important Questions (JNTUA pattern)

- Explain the mechanism of swelling in expansive soils. List the factors affecting swelling.
- Describe the problems caused by expansive soils to buildings, pavements and canals.
- Explain the tests used for identification of expansive soils. How is the free swell index determined?
- Explain the methods of determining the swelling pressure of a soil with a neat sketch.
- Discuss the methods of improvement of expansive soils. Explain the CNS layer concept and lime stabilisation.
- What are the foundation techniques used in expansive soils? Explain the isolation techniques.
- Explain under-reamed piles: geometry, construction and design. Why are they suitable in black cotton soil?